

CLASSIFICATION

1. SVM

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[[82  3]
 [26 23]]
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              precision    recall  f1-score   support

0               0.76         0.96         0.85         85
1               0.88         0.47         0.61         49

 accuracy               0.78         134
  macro avg              0.82         0.72         0.73         134
 weighted avg            0.81         0.78         0.76         134
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Question	Evaluation Metric	Value
What is the percentage of correct classification of both purchase & Non-Purchase, to the total input of the test set ?	Accuracy	0.78
What is the percentage of correct classification of Purchase, to the total Purchase in the test set ?	Recall - Purchase	0.47
What is the percentage of correct classification of NonPurchase, to the total Non-Purchase in the test set ?	Recall - Non Purchase	0.96
What is the percentage of correct classification of Purchase to the sum of correctly classified as Purchase and wrongly classified as Non-Purchase in the test set ?	Precision - Purchase	0.88
What is the percentage of correct classification of Non-Purchase to the sum of correctly classified as Non-Purchase and wrongly classified as Purchase in the test set ?	Precision - Non Purchase	0.76
What is the overall performance of Purchase ?	F1 - Purchase	0.61
What is the overall performance of Non-Purchase ?	F1 - Non Purchase	0.85

	Macro Average	Weighted Average
Precision	0.82	0.82
Recall	0.72	0.78
F1-Measure	0.73	0.76

2. DECISION TREE

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[[76  9]
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precision    recall  f1-score   support

0           0.90      0.89      0.90         85
1           0.82      0.84      0.83         49

accuracy          0.87         134
macro avg          0.86      0.87      0.86         134
weighted avg          0.87      0.87      0.87         134
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Question	Evaluation Metric	Value
What is the percentage of correct classification of both purchase & Non-Purchase, to the total input of the test set ?	Accuracy	0.87
What is the percentage of correct classification of Purchase, to the total Purchase in the test set ?	Recall - Purchase	0.84
What is the percentage of correct classification of NonPurchase, to the total Non-Purchase in the test set ?	Recall - Non Purchase	0.89
What is the percentage of correct classification of Purchase to the sum of correctly classified as Purchase and wrongly classified as Non-Purchase in the test set ?	Precision - Purchase	0.82
What is the percentage of correct classification of Non-Purchase to the sum of correctly classified as Non-Purchase and wrongly classified as Purchase in the test set ?	Precision - Non Purchase	0.90
What is the overall performance of Purchase ?	F1 - Purchase	0.83
What is the overall performance of Non-Purchase ?	F1 - Non Purchase	0.90

	Macro Average	Weighted Average
Precision	0.86	0.87
Recall	0.87	0.87
F1-Measure	0.86	0.87

3. RANDOM FOREST

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[[78  7]
 [ 4 45]]
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precision    recall  f1-score   support

0           0.95      0.92      0.93        85
1           0.87      0.92      0.89        49

accuracy          0.92        134
macro avg         0.91      0.92      0.91        134
weighted avg      0.92      0.92      0.92        134
```

Question	Evaluation Metric	Value
What is the percentage of correct classification of both purchase & Non-Purchase, to the total input of the test set ?	Accuracy	0.92
What is the percentage of correct classification of Purchase, to the total Purchase in the test set ?	Recall - Purchase	0.92
What is the percentage of correct classification of NonPurchase, to the total Non-Purchase in the test set ?	Recall - Non Purchase	0.92
What is the percentage of correct classification of Purchase to the sum of correctly classified as Purchase and wrongly classified as Non-Purchase in the test set ?	Precision - Purchase	0.87
What is the percentage of correct classification of Non-Purchase to the sum of correctly classified as Non-Purchase and wrongly classified as Purchase in the test set ?	Precision - Non Purchase	0.95
What is the overall performance of Purchase ?	F1 - Purchase	0.89
What is the overall performance of Non-Purchase ?	F1 - Non Purchase	0.93

	Macro Average	Weighted Average
Precision	0.91	0.92
Recall	0.92	0.92
F1-Measure	0.92	0.92